



Department of Applied Computing and Artificial Intelligence

Faculty of Computing

SECR1213 NETWORK COMMUNICATIONS

Project Task 3

Choosing The Appropriate LAN Devices (10%)

Group of *Power Rangers*

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3.1 Introduction to The Required Devices

1. Router

A router is a networking device that connects the Faculty of Computing's network to the internet and directs data traffic between different subnets within the building. It allows communication between the labs, the hybrid classroom, the student lounge, and external networks (video conferencing f virtual project meeting).

2. Switch

A switch is a device used to connect multiple devices within the same network (e.g., workstations in a lab) by providing dedicated bandwidth to each device. In this case, switches will connect the PCs, servers, and other equipment in the labs and student lounge to the faculty's internal high-speed network

3. Wireless Access Point

This is a networking device that provides wireless connectivity to the network. It will be used throughout N28b to allow students and staff to connect to the network.

4. Personal Computer (PC)

PCs are the main devices in the labs, used by students for general-purpose tasks, Cisco networking simulations, or working with embedded systems. Each lab will have 30 PCs for the specific needs of the lab.

5. Monitor

Monitors are visual display devices connected to PCs. They are essential for students and staff to interact with software, view data, or participate in video conferencing sessions. Each workstation in the labs will have a monitor.

6. Server

A server provides centralized services such as data storage, software hosting, and network management. It will support activities such as managing lab resources, hosting applications for teaching, and maintaining network security and reliability.

7. Firewall

A firewall is a security device that monitors and controls incoming and outgoing network traffic based on established security rules. It will protect the faculty's network from unauthorized access, ensuring secure communication within the building and with external networks.

8. Modem

A modem is a device that converts digital signals from the faculty's network into analog signals for internet transmission (and vice versa). It connects the faculty's network to the internet service provider, allowing internet-based activities such as virtual meetings.

9. Ethernet Cable

Ethernet cables are wired connections used to link devices like PCs, switches, and routers to the network. They provide high-speed and reliable connections in the labs and other areas of the building where wired connections are necessary.

3.2 Table of Devices

1. Networking Devices and PCs

Device	Specification	Price per unit (RM)	Unit	Total Price (RM)
Router 	Cisco Catalyst 8200 Series Edge • Supports up to 10 Gbps aggregate throughput • Advanced routing capabilities with SD-WAN support • Integrated security with Cisco Umbrella and Zone-Based Firewall • Multiple WAN and LAN ports for scalability and redundancy	10674.57	2	21,349.14
Switch 	Cisco WS-C3650-12X48UZ-L • 48 x Gigabit Ethernet ports with UPOE (Universal Power over Ethernet) support. • 12 x Multigigabit ports supporting speeds of 100 Mbps, 1 Gbps, 2.5 Gbps, 5 Gbps, and 10 Gbps. • 4 x 10-Gigabit Ethernet uplink ports. • Switching capacity: Up to 320 Gbps. • Forwarding rate: Up to 190.48 Mpps. • Supports up to 32K MAC addresses.	5855.33	9	52,697.97

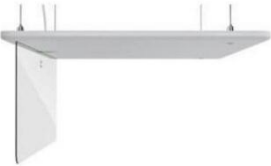
Device	Specification	Price per unit (RM)	Unit	Total Price (RM)
Server 	Cisco C240 M5 Rack Server • Supports up to 2x Intel Xeon Scalable processors (LGA3647 socket). • Options for Bronze, Silver, Gold, and Platinum CPU series with up to 28 cores per CPU. • Up to 24 DIMM slots, supporting up to 6TB of total memory • Dual redundant hot-swappable PSUs (power supply units) for enhanced reliability • Integrated dual 10GBase-T Ethernet ports	1316.01	2	2,632.02
Wireless Access Point 	RUCKUS R610 Enterprise Router • Dual-band 802.11ac Wave 2 enterprise access point • Supports 2.4 GHz and 5 GHz bands with MU-MIMO (Multi-User, Multiple Input, Multiple Output) • Up to 1,900 Mbps combined throughput (800 Mbps on 2.4 GHz and 1,300 Mbps on 5 GHz) • Supports up to 512 concurrent client connections	390.00	17	6,630.00

Device	Specification	Price per unit (RM)	Unit	Total Price (RM)
Firewall 	Huawei USG6355E-AC Enterprise AI Firewall - Firewall throughput: Up to 4 Gbps. - Next-Generation Firewall (NGFW) capabilities, including intrusion prevention, antivirus, data leak prevention, bandwidth management, anti-DDoS, URL filtering, and anti-spam functions. - Application identification and control for over 6,000 applications.	15112.21	1	15,112.21
Desktop (ESL) 	Customize Desktop - CPU: (Intel Core i7-13700K) - GPU: (NVIDIA GeForce RTX 3060 Ti) - Motherboard: (ASUS ROG Strix Z690-A Gaming) - RAM: (Corsair Vengeance DDR4 32GB) - Storage: (Samsung 970 EVO Plus 1TB NVMe SSD) - PSU: (Seasonic Focus GX 750W) - Operating System: (Windows 11 Pro)	7800.00	32	230,400

Device	Specification	Price per unit (RM)	Unit	Total Price (RM)
Desktop (CNL) 	Customize Desktop • CPU: Intel Core i5-13600K • (NVIDIA GeForce RTX 3060) • Motherboard: ASUS TUF Gaming Z690-Plus Wi-Fi D4 • RAM: Corsair Vengeance DDR4 16GB (8GB x 2) 3200MHz • Storage: Kingston NV2 1TB NVMe SSD • PSU: Corsair CV650, 650W 80+ Bronze • Operating System: Windows 11 Pro (OEM)	6,000.00	60	360,000
Desktop (GL, HC) 	Customize Desktop • Intel Core i5-12400F offers great price-to-performance • GTX 1650 provides entry-level gaming performance at a budget-friendly price. • 16GB RAM and 1TB SSD give you good multitasking and storage capabilities. • The B660 motherboard is a cost-effective choice with solid features for a budget system.	3,000.00	24	72,000

Device	Specification	Price per unit (RM)	Unit	Total Price (RM)
Monitor 	Acer Nitro 24" FHD 1MS 180 Hz • Connectivity: HDMI, DisplayPort (Check for the exact input configuration based on the model). • Design: Sleek and modern, with a thin bezel for a clean aesthetic. • Adaptive Sync: Supports AMD FreeSync, which helps eliminate screen tearing during gaming.	519.00	116	61,364
Projector 	Sony VPL-SX536M • Resolution: XGA (1024x768) – suitable for standard office presentations and basic home entertainment. • Brightness: 3,600 lumens – provides bright, clear images even in well-lit rooms, making it ideal for both office environments and home use with ambient light. • Lamp Life: 10,000 hours in eco-mode – offering longer-lasting durability, reducing the need for frequent lamp replacements.	858.05	3	2,574.18

Device	Specification	Price per unit (RM)	Unit	Total Price (RM)
Speaker 	Yamaha VXC3FW Ceiling speaker 3.5" full-range low-profile ceiling speaker (PAIR) • Discreet and clean look with a flush-mount design. • High-quality full-range audio ideal for speech clarity and light music playback. • Simple installation and wide sound dispersion. • Reliable commercial-grade performance.	1,380.00	4	5,520.00
Printer 	HP M605dn LaserJet Enterprise • Up to 58 pages per minute (ppm) for letter-size paper. • First page out in as fast as 7 seconds. • Up to 1200 x 1200 dpi for sharp, professional-quality prints • Standard: 500-sheet input tray and 100-sheet multipurpose tray. • 500-sheet output bin.	1,906.00	11	21966.00

Device	Specification	Price per unit (RM)	Unit	Total Price (RM)
Microphone 	Cisco TelePresence Ceiling Microphone • 180° directional coverage, effectively capturing audio within a 14-foot horizontal and 8-foot vertical range • Frequency response ranges from 80 Hz to 20 kHz, ensuring clear and natural sound reproduction • Signal-to-noise ratio of at least 72 dB, minimizing background noise and enhancing audio clarity.	4,101.12	6	24606.72

2. Connecting Medium

Device	Description	Price per unit (RM)	Unit	Total Price (RM)
Ethernet Cable 	Ethernet Cable CAT6a (100) • 10 Gigabit connections • up to 550MHz frequency and 10GBase-T Ethernet networks • stable and interference-free connections	1499.00	144	215,856

Device	Description	Price per unit (RM)	Unit	Total Price (RM)
Patch panel (48 port) 	FS S3410-48TS-P - Up to 48 PoE+ Ports, Total 740W - Broadcom BCM56150 Chip, Support 4 Units Stacking 1+1 Hot-swappable Power Supplies, Built-in Smart Fans - Cat5e, Cat6, Cat6A Ethernet cables compatibility 2x 10Gb SFP+ Uplinks and 2x Combo SFP Ports	5,877.00	9	52,893.00
Fibre optic cable 	OM4 Multimode Fiber (100m) - Bend Insensitive Fiber & 7.5mm Min. Bend Radius - Precision Zirconia Ferrule Connectors Ensure Low Loss - Zipcord Reinforced Fiber and Reduces Crimping Risk	2000.00	18	36,000

Device	Description	Price per unit (RM)	Unit	Total Price (RM)
Cable management rack 	Iron plate + plastic cable management rack • Arranging jumpers to ensure that jumpers do not look too messy. • regulate the jumper at the front end of the distribution frame to make the front-end jumper more regular and orderly • Cover the combed wires in the wire board, and the cables inserted in the wire rack are not visible on the surface, so the appearance is neat and orderly	16.27	2	32.54
Wall panel 	Keystone Jack Network Faceplate • 8-position 8-conductor (8P8C) and designed for computer networking • 568A and 568B colour wiring diagram is integrated on the lid • The modular plugs are manufactured from high-impact clear polycarbonate to withstand everyday wear and tear	3.50	116	406.00

Total for all devices: **RM1,182,039.78**

Reflection

1. Are you surprised by the prices? How were you surprised?

Yes, the prices are surprising. They are higher than expected for basic features or much lower for the devices with more advanced functionalities. It is surprising when a well-known brand is very expensive compared to other brand or a less popular brand that provides similar functionalities at a much lower price.

2. Have you ever considered cost as a factor for choosing networking devices?

Yes, because cost is an important factor whilst choosing the networking devices. We have to make sure that the cost of devices fit to the budget whilst still meeting the needs. The cheaper options may not have all the features that we need, but the more expensive devices might provide more than the required features. Therefore, finding the right balance between cost and the functionalities is important.

3. What are the major differences between the same devices from different brands?

For example, Cisco and Huawei routers.

I. Router

Brand	Cisco Catalyst 8200 Series Edge	Huawei AR8000
Operating System	Cisco IOS XE	VRP
SD-WAN Capabilities	Strong SD-WAN features with cloud-based vManage	Competitive SD-WAN with Agile Controller
Security Features	Advanced threat intelligence, AI-driven malware detection.	Comprehensive security with hardware firewalls and strong encryption
Cost and Licensing	Higher upfront and ongoing costs with tiered DNA licenses.	Generally more affordable hardware and simpler licensing.

The major difference is that the Cisco Catalyst 8200 Series Edge offers a more advanced operating system (IOS XE), superior SD-WAN capabilities with extensive cloud-based management (vManage), and AI-driven threat detection, making it ideal for high-performance enterprise networks and security-critical environments. In contrast, the Huawei AR8000 Series provides cost-effective solutions with a user-friendly VRP operating system, competitive SD-WAN capabilities, and strong hardware encryption, making it suitable for businesses seeking affordable and efficient WAN deployments.

II. Switch

Brand	Cisco WS-C3650-12X48UZ-L	Huawei S6720-50L-HI-48S-AC
Port Configuration	48 Multigigabit Ethernet ports, 12 x 10G SFP+ uplinks	48 x 10G SFP+ ports, 6 x 40G QSFP+ uplinks.
PoE Capability	Supports Cisco UPOE, up to 60W per port.	No PoE support; focused on high-density fibre ports.
Performance	176 Gbps forwarding bandwidth, 448 Gbps switching capacity	2.56 Tbps switching capacity, optimized for high-density applications
Wireless Integration	Integrated wireless controller supports 25 APs and 1000 clients.	No wireless controller integration.

The major difference is that the Cisco WS-C3650-12X48UZ-L excels in PoE support and wireless integration, making it ideal for branch offices or enterprises needing power for connected devices and unified wired-wireless management. In contrast, the Huawei S6720-50L-HI-48S-AC offers superior switching capacity and high-density 10G and 40G fibre connectivity, catering to high-bandwidth data centre and aggregation scenarios

III. Wireless Access Point

Brand	Ruckus R610	Cisco Catalyst 9105AXI
Wireless Standard	Wi-Fi 5 (802.11ac Wave 2)	Wi-Fi 6 (802.11ax)
Client Capacity	Up to 512 clients	Up to 200 clients per radio
Antenna Technology	BeamFlex+ adaptive antennas for signal optimization and interference mitigation	Internal omnidirectional antennas with optimized coverage for Wi-Fi 6
Power Options	Supports PoE (802.3af) for simplified installation	Supports PoE+ (802.3at) for additional power and future scalability

The major difference is that the Cisco Catalyst 9105AXI is a Wi-Fi 6 device offering higher efficiency, better client management in dense environments, and support for PoE+, making it ideal for future-ready deployments. In contrast, the Ruckus R610 leverages BeamFlex+ technology to optimize performance and interference mitigation in high-density Wi-Fi 5 networks, offering a cost-effective solution for enterprises with current-generation needs.

IV. Personal Computer (PC)

Brand	Dell Inspiron Desktop	HP Pavilion Desktop
Processor	Intel Core i5 (12 th Gen)	AMD Ryzen 5
Storage	1TB HDD + 256 SDD	512GB SDD
Graphic Card	Integrated Intel UHD Graphics	Integrated AMD Radeon Graphics
Connectivity	WiFi-6, Bluetooth 5.2	WiFi-5, Bluetooth 4.2

The major difference is Dell Inspiron Desktop provides a dual-storage solution. It provides 1TB HDD and 256 SDD which makes it suitable for users who need large storage capacity and faster boot times. Additionally, Dell provides better connectivity with WiFi-6 and Bluetooth 5.2 for faster and more stable connections. HP Pavilion with AMD Ryzen 5 processor and AMD Radeon Graphics is optimised for better graphics performance, but provides less storage capacity and older connectivity standards.

V. Monitor

Brand	Acer Nitro XV240Y M3	Phillips 24E2FE
Price (RM)	449	539
Refresh Rate	180 Hz	75 Hz
Connectivity	HDMI, DisplayPort	HDMI, VGA
Resolution	Full HD (1920x1080)	Full HD (1920x1080)

The major difference is about the refresh rate. The Acer Nitro XV240Y M3 provides a high 180 Hz refresh rate. It is suitable for gaming and fast-paced activities, whilst the Phillips 24E2FE has a lower 75 Hz refresh rate, only suitable for light and general use. The Acer monitor is also more affordable and only cost RM449. It offers both HDMI and DisplayPort connectivity. In contrast, the Phillips 24E2FE lacks of high refresh rate, but with higher price.

VI. Server

Brand	Cisco C240 M5 Rack Server	Dell PowerEdge R740
Form Factor	2U	2U
Memory Capacity	Up to 6 TB DDR4 (24 DIMM slots)	Up to 3 TB DDR4 (24 DIMM slots)
GPU Support	Supports up to 6 GPUs	Supports up to 3 GPUs
Management Tools	Cisco Intersight, UCS Manager	iDRAC9 with Lifecycle Controller

The major difference is that the Cisco C240 M5 Rack Server provides higher memory capacity (up to 6 TB) and greater GPU support (up to 6 GPUs), making it ideal for demanding workloads like AI, ML, and big data analytics. In contrast, the Dell PowerEdge R740 focuses on balanced performance and ease of management with iDRAC9, which is well-suited for general-purpose workloads and enterprise environments.

VII. Patch Panel

Brand	FS S3410-48TS-P	TP-Link T2600G-28TS
Form Factor	1U Rackmount	1U Rackmount
Forwarding Rate	132 Mpps	41.66 Mpps
Power Supply	Dual power supply	Single power supply
Power Budget	740W	192W

The major difference is that the FS S3410-48TS-P offers a higher forwarding rate (132 Mpps) and a significantly larger power budget (740W), making it suitable for high-performance and PoE-intensive applications like enterprise-level deployments. In contrast, the TP-Link T2600G-28TS is designed for smaller-scale environments with lower forwarding and power requirements, offering a cost-effective solution for basic networking needs.

VIII. Fibre Optic

Brand	OM4 Multimode Fibre (100m)	OM3 Multimode Fibre (100m)
Core Diameter	50 µm	50 µm
Jacket Colour	Magenta (for easy identification)	Aqua (for easy identification)
Application	High-speed data centers, cloud computing	General data center applications

The major difference is that OM4 Multimode Fibre provides significantly higher bandwidth, making it ideal for high-speed applications like data centres and cloud computing that require faster data transmission over longer distances. In contrast, OM3 Multimode Fibre is suited for general data centre applications where slightly lower bandwidth requirements and shorter distances are typical.

XI. Wall Panel

Brand	Keystone Jack Network Faceplate	Tripp Lite N080-101 1-Port Keystone Wallplate
Installation Type	Wall-mounted	Wall-mounted
Type	Single Port Network Faceplate	Single-Gang Universal Keystone Wallplate
Port Compatibility	RJ45, Cat5e, Cat6, multimedia interfaces	Compatible with any keystone jack (Cat5e/6/6a, USB, HDMI, etc.)
Material	Anti-burn PC plastic	High-impact polycarbonate

The major difference is that the Keystone Jack Network Faceplate is specifically designed for networking applications, supporting RJ45, Cat5e, and Cat6 connections, while the Tripp Lite N080-101 offers broader compatibility, supporting a variety of keystone jacks (including USB, HDMI, and Cat5e/6/6a), making it more versatile for different multimedia and data connections. Additionally, the Tripp Lite wall plate is made from high-impact polycarbonate, known for durability, while the Keystone Jack Network Faceplate uses anti-burn PC plastic, providing added fire safety.

Meeting Minutes #1

Date/Time		20 th December 2024 5:00pm	
Location		Online (Google Meet)	
Agenda		1. Understand the details about Task 3	
		2. Discuss the materials prepared in advance	
		3. Determine possible network devices to be included in the project and the quantity and price for them	
		4. Task distribution for all members	
Meeting Host		Brendan Chia Yan Fei	
Attendance			
Name		Time	Reason For Absence
Elijah She Yu Sheng		5:00pm	-
Lau Yan Kai		5:00pm	-
Brendan Chia Yan Fei		5:00pm	-
Chew Chiu Xian		5:00pm	-
Minutes			
No.	Item Discussed	Details	Person-In-Charge
1	Router	- Discussed Cisco Catalyst 8200 Series Edge router. It supports up to 10 Gbps throughput, advanced routing with SD-WAN, security features like Cisco Umbrella.	Elijah
2	Switch	- Cisco WS-C3650-12X48UZ-L Switch, with 48 Gigabit ports, 12 Multigigabit ports, and PoE support. It offers high switching capacity and bandwidth for lab needs	Brendan

3	Wireless Access Point	- Ruckus R610 Access Point supports 802.11ac Wave 2, 1,900 Mbps throughput, and 512 client capacity. To be used throughout N28b for wireless connectivity.	Yan Kai
4	Personal Computer (PC)	- Discussed specifications of various PCs, with 32 customized desktops for ESL, 60 for CNL, and 24 for GL and HC labs. They have Intel Core i5, RAM, SSD, GPU.	Elijah
5	Monitor	- Acer Nitro 24" FHD 1ms 180Hz monitors for each lab workstation. To enhance student experience with smooth visuals for work and video conferencing.	Yan Kai
6	Server	- Cisco C240 M5 Rack Server to support up to 2x Intel Xeon CPUs, large memory, and network services. Critical for managing lab resources and applications.	Yan Kai
7	Firewall	- Huawei USG6355E-AC Firewall with 4 Gbps throughput, offering advanced security features like intrusion prevention, data leak prevention, and anti-DDoS	Elijah
8	Modem	- Modem to convert digital signals for internet transmission, ensuring smooth connectivity for virtual meetings and online activities.	Brendan
9	Ethernet Cable	- Ethernet CAT6a cables for 10Gbit connections, required for reliable wired connections across labs and other areas	Elijah
10	Patch Panel	- FS S3410-48TS-P 48 port patch panel, with 740W power budget. Key for organizing network connections and cables	Brendan
11	Fibre Optic Cable	- OM4 Multimode Fibre for high-speed data transmission in data centers. Suitable for cloud computing and high-speed applications.	Chiu Xian
12	Cable Management	- Cable management racks for organizing cables in the distribution frame to ensure neat and ordered wiring	Chiu Xian
13	Wall Panel	- Keystone Jack Network Faceplates for networking, providing a clean installation for Ethernet connections and better organization of cables	Chiu Xian

14	Reflection on Costs	- Discussed the varying costs of networking devices. Mentioned the importance of balancing cost with device functionality for budget compliance.	Brendan
15	Major Differences in Brands	- Discussed key differences in routers, switches, PCs, servers, and other devices from Cisco, Huawei, and other brands, including performance and cost factors.	Yan Kai

--- END OF DOCUMENTATION ---